

Name: _____

A Level Mathematics : Pure Mathematics

Topic 6 :Exponentials and logarithms -Use of logarithmic graphs to estimate parameters

Date:

Time:

Total marks available:

Total marks achieved: _____

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Questions

Q1.

(a) Given that

$$\frac{x^2 + 8x - 3}{x + 2} \equiv Ax + B + \frac{C}{x + 2} \quad x \in \mathbb{R} \quad x \neq -2 \quad (3)$$

find the values of the constants A , B and C

(b) Hence, using algebraic integration, find the exact value of

$$\int_0^6 \frac{x^2 + 8x - 3}{x + 2} dx \quad (4)$$

giving your answer in the form $a + b \ln 2$ where a and b are integers to be found.

(Total for question = 7 marks)

Q2.

(a) Use the substitution $x = u^2 + 1$ to show that

$$\int_5^{10} \frac{3 dx}{(x-1)(3+2\sqrt{x-1})} = \int_p^q \frac{6 du}{u(3+2u)}$$

where p and q are positive constants to be found.

(4)

(b) Hence, using algebraic integration, show that

$$\int_5^{10} \frac{3 dx}{(x-1)(3+2\sqrt{x-1})} = \ln a$$

where a is a rational constant to be found.

(6)

(Total for question = 10 marks)

In a controlled experiment, the number of microbes, N , present in a culture T days after the start of the experiment were counted.

N and T are expected to satisfy a relationship of the form

$$N = aT^b, \quad \text{where } a \text{ and } b \text{ are constants}$$

(a) Show that this relationship can be expressed in the form

$$\log_{10} N = m \log_{10} T + c$$

giving m and c in terms of the constants a and/or b .

(2)

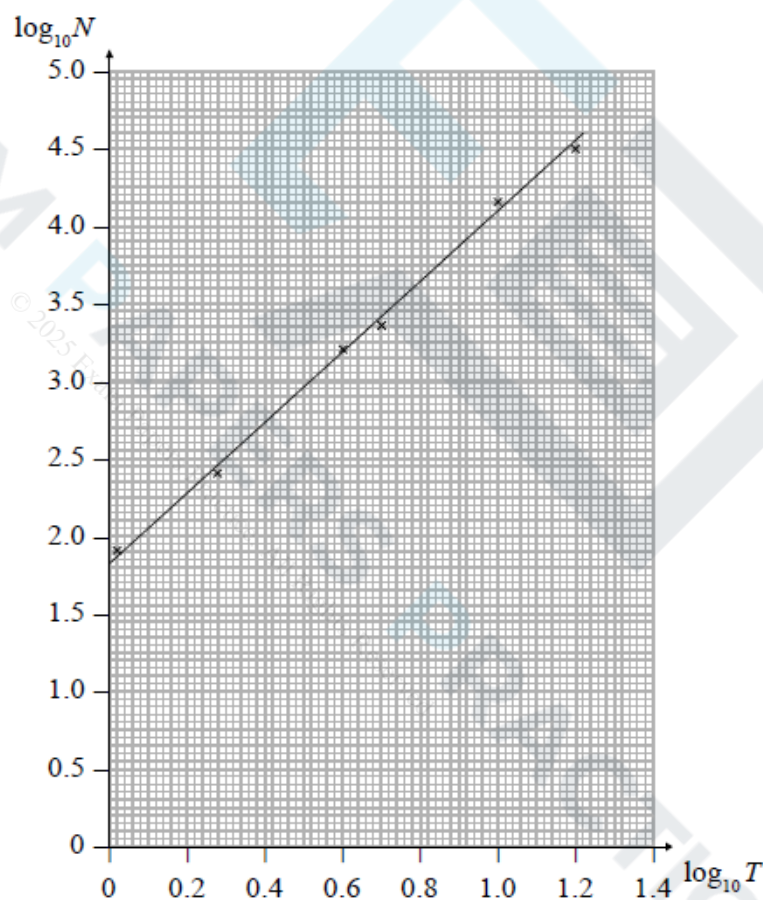


Figure 3

Figure 3 shows the line of best fit for values of $\log_{10} N$ plotted against values of $\log_{10} T$

(b) Use the information provided to estimate the number of microbes present in the culture 3 days after the start of the experiment.

(4)

(c) Explain why the information provided could not reliably be used to estimate the day when the number of microbes in the culture first exceeds 1 000 000.

(2)

(d) With reference to the model, interpret the value of the constant a .

(1)

(Total for question = 9 marks)

Q4.

By taking logarithms of both sides, solve the equation

$$4^{3p-1} = 5^{210}$$

giving the value of p to one decimal place.

(Total for question = 3 marks)

Q5.

(a) Given that

$$2 \log (4 - x) = \log (x + 8)$$

show that

$$x^2 - 9x + 8 = 0$$

(3)

(b) (i) Write down the roots of the equation

$$x^2 - 9x + 8 = 0$$

(ii) State which of the roots in (b)(i) is **not** a solution of

$$2 \log (4 - x) = \log (x + 8)$$

giving a reason for your answer.

(2)

(Total for question = 5 marks)